



BITS Pilani
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Professional AI/ML Programme

AWS Model Engineering Lab

Lab 1: Build & Register · Lab 1.1: Deploy & Operationalize

Duration: 2.5 Hours | BITS Pilani Digital

Session at a Glance

2.5 hrs

Total Duration

2 Labs

Lab 1 + Lab 1.1

17 Steps

End-to-End Flow

26 Artifacts

Produced in S3

6 AWS Services

S3 · SM · EventBridge

Setup

Environment config, S3 paths, helper utilities

Lab 1

Steps 1–9: Freeze → Splits → EDA → Pipeline → Selection → Bakeoff → Registry → Model Card

Break

Q&A buffer

Lab 1.1

Steps 1–8: Validate → Approve → Package → Deploy → Schedule → Events → Cluster

Wrap-up

26-artifact audit, handoff checklists, go-live gate

Lab 1 — Build & Register a Candidate

Steps 1–9 · ~85 minutes · S3 · SageMaker Processing · Training · Model Registry

1 Freeze Dataset Immutable snapshot schema · manifest MD5 hash 10m	2 Split Contract 60/20/20 stratified 5-fold CV time/group patterns 5m	3 EDA & Leakage Correlation · ID- like missingness · skew cardinality 10m	4 Feature Pipeline ColumnTransformer Winsoriser · TF- IDF 3 YAML configs 15m	5 Feature Selection MI · LASSO · RF permutation PCA artifact 10m	6 Model Bakeoff LogReg / RF / GBM 6 artifacts/run visual comparison 20m	7 Evaluate & Pick AUC gap penalty CV variance calibration 10m	8 Register Winner Model Package Group lineage metadata PendingApproval 6m	9 Model Card 8 governance sections · intended use · limitations 1m
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Steps 1–3: Dataset Freeze · Split Contract · Leakage Scan

Everything that must happen before fitting — in order

Step 1 — Freeze the Dataset (10 min)

- Upload raw parquet to S3 — zero transformations
- Compute MD5 hash → attach to manifest.json
- manifest.json: version, row count, timestamp, data quality summary
- Rule: if it changes, it is a new version — never overwrite

Step 2 — Split Contract (5 min)

- Stratified 60/20/20 split — class balance verified across all splits
- 5-fold CV fold definitions for selection stability
- Time-based split: quantile cutoffs on date column
- Group-based split: GroupShuffleSplit on customer_id
- Persist as parquet — never recompute between runs

Step 3 — EDA & Leakage Scan (10 min)

- Correlation with target > 0.85 → leakage risk flag
- ID-like columns ($>95\%$ unique) → likely row identifiers
- Near-constant features ($\text{std} < 1e-5$) → zero information
- High-missingness ($>30\%$) → imputation strategy decision
- High cardinality categoricals (>20 unique) → encoding choice
- High skew ($|\text{skew}| > 1.5$) → log/power transform candidate
- Shuffle-target test (Step 7):
 - Shuffle target labels → recompute AUC 30 times
 - Real AUC – shuffle mean AUC < 0.05 → WARNING
- 5 EDA artifacts written to S3:
 - eda_summary.md · leakage_risks.json
 - feature_profile.parquet · high_cardinality_columns.json
 - transformation_candidates.json

Steps 4–5: Feature Pipeline · Selection · Reduction

Fit on train only — persist the pipeline — rank by multiple methods



Step 4 — Feature Pipeline (15 min)

- Numeric: impute(median) → StandardScaler
- Categorical: impute('MISSING') → OrdinalEncoder(unknown=-1)
- Optional Winsorizer: clip at 1st/99th percentile
- Optional TF-IDF: for text columns in pipeline
- Fit on training split ONLY — transform all three splits
- Persist as preprocess_v1.joblib
- 3 Feature Configs (YAML):
 - baseline_v1 → all output features
 - selected_v1 → MI + RF + LASSO top-20
 - reduced_v1 → PCA 15 components
- Feature Store: use FeatureGroup.ingest() when features need cross-model reuse or real-time lookup

Step 5 — Feature Selection (10 min)

- Filter: mutual_info_classif + VarianceThreshold(0.01)
- Embedded: LassoCV(5-fold) — zero-coeff features removed
- Embedded: RandomForest importances (100 estimators)
- Average rank of MI + RF rankings → top-20 final list
- Permutation importance on val set: 5 repeats → mean ± std
- stability_score = 1 - mean(std / |mean_importance|)
- 4 artifacts written to S3:
 - selected_features_v1.json
 - selection_stability_v1.json
 - feature_impact_report_v1.md
 - pca_v1.joblib (already fitted in Step 4)

Step 6 — Model Bakeoff

3 candidates · same split contract · same feature-config IDs · 6 artifacts per run (1 20 min)

Linear Baseline

Logistic Regression

```
feat_config: baseline_v1
```

```
run_id: run_001
```

- C=1.0, max_iter=500
- All features (baseline config)
- Minimum viable performance floor
- Fast · interpretable · calibrates well
- Establishes comparison anchor

Tree Ensemble

Random Forest

```
feat_config: selected_v1
```

```
run_id: run_002
```

- 200 estimators, max_depth=8
- Top-20 selected features
- Handles non-linearity & interactions
- Robust to outliers
- Strong permutation importance source

Boosted Ensemble

GradientBoosting

```
feat_config: reduced_v1
```

```
run_id: run_003
```

- 150 trees, lr=0.05, depth=4
- PCA-reduced features
- Alt: SVC(probability=True)
- Alt: GaussianNB for sparse data
- Higher bias but regularised

Per-run artifacts (×3): params.json · metrics.json · feature_config.json · env.json · model.tar.gz · inference_spec.json

Steps 7–9: Evaluate · Register · Model Card

Standardised scoring → Registry governance → Model card documentation

Step 7 — Standardised Evaluation (10 min)

- $\text{score} = \text{val_AUC} - 0.3 \times (\text{train_AUC} - \text{val_AUC})$
- CV variance: 5-fold std > 0.03 → investigate
- Model size: joblib dump → measure KB
- Calibration: `calibration_curve()` → Brier score
- Shuffle-target test (30 trials): $\Delta > 0.05 = \text{PASS}$
- Threshold: `precision_recall_curve` → max-F1

- 4 artifacts: `comparison.json`
- `model_bakeoff_v1.md`
- `shuffle_target_test.json`
- `calibration_report.json`

Step 8 — Model Registry (6 min)

- `create_model_package_group` for project
- Register: `InferenceSpec` + `ModelApprovalStatus`
- = `PendingManualApproval` (never auto-approve)
- Lineage: `dataset_version` · `split_version`
- `feature_config_id` · `commit_SHA`
- Artifact: `registry/candidate_submission_v1.json`

Step 9 — Model Card (1 min)

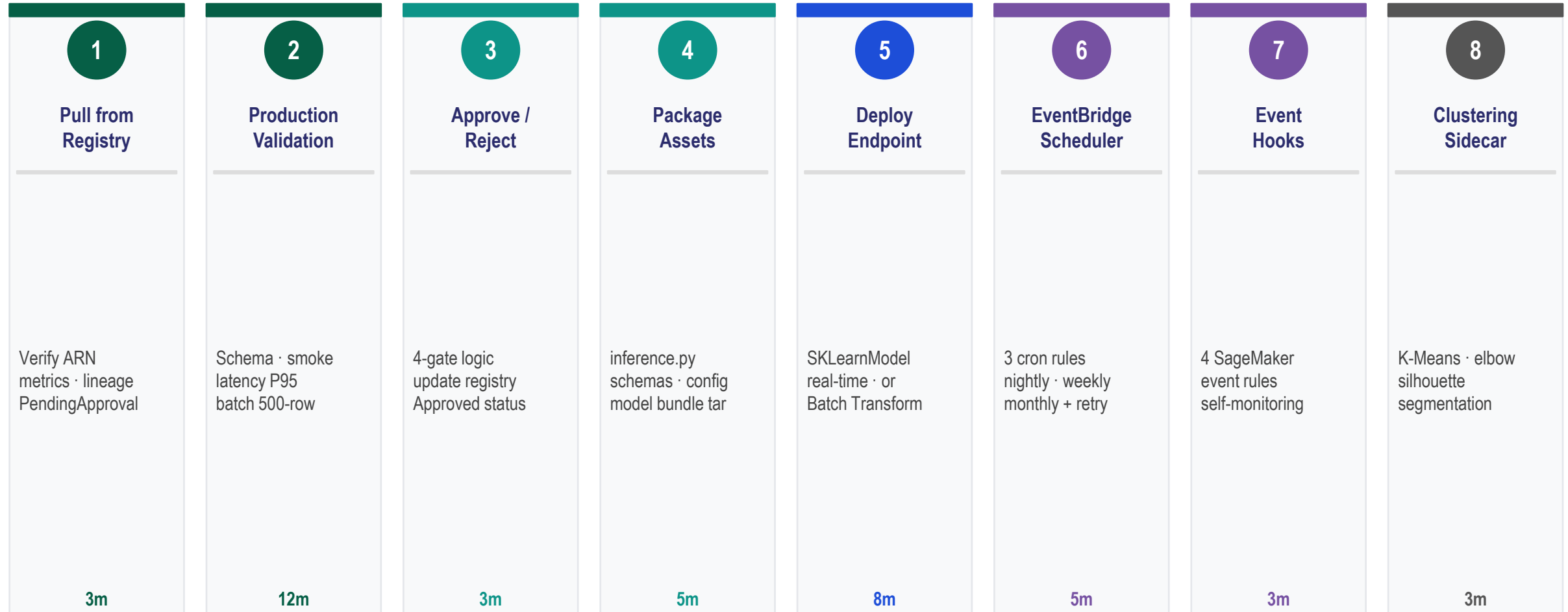
- 8 required sections:
- 1. Intended use + out-of-scope uses
- 2. Model details (algorithm, version)
- 3. Performance metrics (train/val/test)
- 4. Training data snapshot
- 5. Leakage check results
- 6. Known limitations
- 7. Fairness / risk notes
- 8. Registry reference (ARN)

Lab 1 Deliverables

- ✓ Registered candidate model
- ✓ Preprocessing pipeline artifact
- ✓ 3 feature configuration YAMLS
- ✓ Model bakeoff report
- ✓ EDA & leakage summary
- ✓ Model card (8 sections)
- ✓ Release recommendation
(Promote / Revise / Reject)

Lab 1.1 — Promote, Deploy & Operationalize

Steps 1–8 · ~45 minutes · Registry · Endpoint · EventBridge



Lab 1.1 Step 2 — Production Validation Suite

7 checks · runtime correctness only · NOT more experimentation (12 min)

 Schema Contract	 E2E Smoke Test	 Latency Benchmark	 Memory Footprint	 Batch Smoke Test	 Threshold Sanity	 Rollback Check
Compare inference_spec.json expected features vs. live payload shape. Type mismatches fail silently.	Load preprocessor.joblib + model.tar.gz together. Transform raw rows → predict. Must run as a unit.	100 single-row predictions → P50/P95/P99 ms. Budget: P95 < 50ms. Establishes SLA baseline.	tracemalloc peak on 1000-row transform. Flag memory spikes before they hit production containers.	Score 500 representative rows end-to-end. No errors, score in [0,1], positive rate reasonable.	precision_recall_curve → max-F1 threshold. Default 0.5 is rarely optimal for imbalanced targets.	Document previous approved ARN in deploy_config.yaml → rollback_model_arn before promotion.

Lab 1.1 Steps 5–7: Deploy · Schedule · Event Hooks

Real-time endpoint · EventBridge Scheduler · EventBridge Rules

Step 5 — Deploy (⌚ 8 min)

- Real-Time Endpoint:
- `SKLearnModel.deploy()` ml.m5.large
- ~\$0.115/hr — always-on cost
- Best: low-latency, user-facing APIs
- Batch Transform (alternative):
- Transformer → pay per job ~\$0.115/run
- Best: nightly churn, weekly reports
- `inference.py`: `model_fn` · `input_fn`
- `predict_fn` · `output_fn`
- Bundle: model + preprocessor in `tar.gz`
- Delete endpoint after session!

Step 6 — EventBridge Scheduler (⌚ 5 min)

- `cron(0 2 * * ? *)` → Nightly batch scoring
- `cron(0 3 ? * MON *)` → Weekly retraining
- `cron(0 4 1 * ? *)` → Monthly drift check
- RetryPolicy: `MaxAttempts=3`, `MaxAge=3600s`
- No retry = stale predictions for 24h!

Step 7 — EventBridge Rules (⌚ 3 min)

- `TrainingJobStatus=Completed` → validation job
- `EndpointStatus=Failed` → SNS → on-call alert
- `ModelApproval=Approved` → CD pipeline trigger
- `PipelineExecution=Failed` → PagerDuty high sev
- Scheduler=time · Rules=event · both needed

Step 8 — Clustering Sidecar (⌚ 3 min)

- K-Means on full preprocessed dataset
- Elbow analysis `K=2..8` → best K by silhouette
- Davies-Bouldin as secondary quality metric
- Cluster profiles: size + target rate per cluster
- 2D PCA projection → PNG for business report
- `cluster_eval.py` → reusable Processing script
- Rule: DO NOT use clusters to gate or override supervised production decision
- Use for: segmentation · anomaly exploration
- · feature candidates for Model v2

Lab 1.1 Steps 3–4: Approve · Package Deployment Assets

Formal approval gate → complete deployment bundle

Step 3 — Approve or Reject (⏱ 3 min)

- 4-gate approval logic:
- schema_passed AND leakage_passed
- AND val_AUC ≥ 0.75 AND P95 latency < 50ms
- update_model_package() → ModelApprovalStatus = Approved
- Approval reason string → creates auditable trail
- Moving to Approved can trigger CD pipeline via EventBridge

Step 4 — Package Assets (⏱ 5 min)

- inference.py: model_fn, input_fn, predict_fn, output_fn
- model_bundle_v1.tar.gz: model + preprocessor + inference.py
- requirements.txt: pinned — no loose version ranges
- input_schema.json + output_schema.json
- deploy_config.yaml: instance, scaling, rollback_model_arn

Deployment Bundle — all versioned together, all in S3:

model_bundle_v1.tar.gz

inference.py

requirements.txt

input_schema.json

output_schema.json

deploy_config.yaml

Go-Live Gate (all 10 must pass before production promotion):

- | | | | | |
|-----------------------------|----------------------|-----------------------|---------------------------|---------------------------|
| ✓ Offline metrics threshold | ✓ P95 latency < 50ms | ✓ Schema validated | ✓ No critical leakage | ✓ Registry Approved |
| ✓ Deployment target chosen | ✓ Retry policy set | ✓ Rollback documented | ✓ Event alerts configured | ✓ Post-deploy sample done |

Handoff Checklists

Lab 1 → Lab 1.1 gate · DS → ML Engineering handoff package

Lab 1 → Lab 1.1 Handoff Gate (14 items)

- | | |
|---|---|
| 1. Dataset version ID + manifest.json verified | 2. Schema file + contract documented |
| 3. Split indices frozen as parquet (train/val/test) | 4. CV fold definitions (cv_folds.json) persisted |
| 5. Preprocessing artifact (preprocess_v1.joblib) | 6. 3 feature config YAMLS (baseline/selected/reduced) |
| 7. Selected features + stability report in S3 | 8. Model artifact (model.tar.gz) in S3 |
| 9. Metrics report (metrics.json) for winning run | 10. Model bakeoff comparison report |
| 11. Leakage test — no critical findings | 12. Model card (model_card_v1.md) linked to registry |
| 13. Registry entry — PendingManualApproval | 14. Training commit SHA in registry metadata |

DS → ML Engineering Handoff Package (8 items)

- | | |
|------------------------------------|--|
| Approved feature contract | selected_v1.yaml — exact feature list, types, encoding rules |
| Exact inference schema | input_schema.json + output_schema.json — no ambiguity |
| Artifact compatibility note | smoke test passed — see validation_report_v1.md |
| Expected latency budget | P50/P95/P99 — see release/latency_smoke.json |
| Expected batch volume / QPS | records per scoring run — see deploy_config.yaml |
| Dependency list | requirements.txt — all packages pinned |
| Rollback model version | ARN in deploy_config.yaml → rollback_model_arn |
| Business + Technical owner | named contacts for escalation and approval |

8 Core Teaching Points

Key concepts learners must walk away knowing

Split contract first

Freeze → splits → EDA → pipeline. Any other order risks leakage. EDA on the full dataset leaks distribution knowledge into preprocessing decisions.

Leakage is silent

Shuffle-target test is the only reliable post-selection leakage catch. Real AUC – shuffle mean < 0.05 is a red flag requiring investigation.

feature_config_id is mandatory

A run without its feature_config_id cannot be reproduced. Comparing candidates is only valid when both logged which features they received.

Registry = governance

PendingManualApproval is not a formality — it creates an audit trail. In regulated industries, that trail may have legal significance.

Model card before deployment

Cards written after deployment are documentation theatre. Known limitations are not a weakness — they are a sign of a responsible practitioner.

Validate, don't experiment

Lab 1.1 validation is about runtime correctness, not metric optimisation. The 7-check suite tests the production artifact, not the research question.

Retry policy is mandatory

A failed nightly scoring job with no retry = stale predictions for 24 hours downstream. RetryPolicy: MaxAttempts=3, MaxEventAge=3600s.

Clustering is a sidecar

Never use clustering output to gate or override the supervised production decision. Use it for segmentation insight and feature generation for Model v2.

Complete S3 Artifact Inventory

All 26+ artifacts produced — verified by wrap-up audit cell · Relative to s3://<bucket>/model-engineering-lab/

Datasets & Splits

- datasets/v1/raw/bank_marketing_raw.parquet
- datasets/v1/schema/schema.json
- datasets/v1/manifest.json
- splits/split_v1/train_idx.parquet
- splits/split_v1/cv_folds.json

EDA Outputs

- eda/eda_summary.md
- eda/leakage_risks.json
- eda/feature_profile.parquet
- eda/high_cardinality_columns.json
- eda/transformation_candidates.json

Pipelines & Configs

- pipelines/preprocess_v1.joblib
- pipelines/pca_v1.joblib
- feature_configs/baseline_v1.yaml
- feature_configs/selected_v1.yaml
- feature_configs/reduced_v1.yaml

Feature Selection

- features/selected_features_v1.json
- features/selection_stability_v1.json
- features/feature_impact_report_v1.md

Training Runs (x3)

- runs/run_00N/params.json
- runs/run_00N/metrics.json
- runs/run_00N/feature_config.json
- runs/run_00N/model.tar.gz
- runs/run_00N/inference_spec.json

Evaluation

- evaluation/comparison.json
- evaluation/model_bakeoff_v1.md
- evaluation/shuffle_target_test.json
- evaluation/calibration_report.json

Registry & Card

- registry/candidate_submission_v1.json
- model_cards/model_card_v1.md

Release & Deploy

- release/validation_report_v1.md
- release/latency_smoke.json
- deploy/model_bundle_v1.tar.gz
- deploy/inference.py
- deploy/deploy_config.yaml

Unsupervised

- unsupervised/cluster_metrics.json
- unsupervised/clustering_brief_v1.md
- unsupervised/cluster_eval.py
- unsupervised/cluster_projection.png

Thank You